

Supplementary Material for From Configurations to Claims: An Evidence Audit of Urban EV Charging Forecasting

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1 Audit steps and wording rules

Table 1: Fail-closed audit steps used in this study. Each transition requires a checkable artifact; a failed condition moves to *blocked* rather than metric interpretation.

Stage	Required state	Minimum record	Fail-closed transition	Supported statement
1. Register	protocol fixed before aggregate evidence	data role, target, split, forecast origin, model/config ID, seed, endpoint, controls, gate, stop rule	missing or mutable field → BLOCKED	registered only
2. Execute	registered configuration	run ID, configuration fingerprint, code/environment identity, attempt state, immutable output paths	failure → terminal failed run; non-run → non-complete	executed or failed ; never reproduced by itself
3. Close	all expected cells terminal	expected/observed cardinality; target, order, time, spatial and mask hashes; prediction hashes; provenance links	incomplete or mismatched family → repair required	closed-unverified
4. Audit	closed matrix; metrics still locked	identity audit, boundary audit, control validity, repair parent/child hashes, prediction-change flag	any defect → invalidate affected formal family; non-selective rerun where selection could bias repair	audit-pass
5. Unlock	every registered audit passes	unlock authority, conditions, receipt hash, access counters, exact audited matrix hash	absent receipt or premature access → BLOCKED	metric-unlocked for the named role only
6. Decide	metric implementation and reference fixed	endpoint, reference, effect, uncertainty, consistency checks, conjunctive gate outcome	any required check fails → versioned hypothesis terminates	gate-pass or gate-fail ; subchecks stay typed
7. Write	decision and evidence chain retained	claim ID, authoritative artifact and hash, evaluation type, allowed wording, prohibited wording, scope limit	missing lineage → claim withheld	scoped manuscript statement

Table 2: Fail-closed audit steps (continued): wording rules that keep diagnostic, proxy, engineering, and procedural results separate from stronger claims.

Evidence type	Required wording	Example permitted wording	Unsupported wording
Oracle diagnostic	[target-dependent role] + [dataset boundary] + [reference] + [opportunity bound]	“indicates an infeasible upper bound”	“is deployable” or “can be recovered”
Proxy prediction	[proxy label] + [forecast-origin information set] + [discrimination/calibration]	“predicts relative expert advantage”	“improves forecast RMSE” or “identifies a cause”
Real-GT development/internal	[data role] + [reference] + [endpoint] + [effect/interval]	“improves internal rolling RMSE by $x\%$ ”	“generalizes”, “blind test”, or “state of the art”
Engineering measurement	[device + workload + precision + batch] + [endpoint] + [comparison]	“measured P50 was x ms on the frozen workload”	“is efficient” without a matching serving study
Conjunctive gate	[candidate] + [all required checks] + [pass/fail] + [authorized next action]	“failed the 1% point-effect requirement; stage stopped”	replacing the frozen question with a positive subcheck
Artifact-integrity event	[defect] + [pre-metric state] + [repair/audit] + [unlock outcome]	“blocked interpretation until identity audit passed”	“repair improved accuracy”
Procedural data boundary	[asset role] + [storage state] + [analytical-use state] + [threat-model limit]	“vault-materialized; analytical access after lock was zero”	“never materialized” or “cryptographically inaccessible”

2 Model identities and local adaptations

Table 3: Model identity, local adaptation, and claim scope. “Unknown/not claimed” is used when the frozen project artifacts do not establish a unique upstream identity or an authoritative field.

Paper label	Frozen implementation identity	Data/input contract and local changes	Role and explicit non-claim
UrbanEV CAPER	Existing exact-target expert predictions; legacy artifact label phase_only . The latency artifact reports 47,992 parameters. Upstream paper/repository checkpoint identity: unknown/not claimed .	UrbanEV panel, 168-hour project context and four horizons in the shared evaluation. The paper reuses predictions and does not reconstruct an unrecorded upstream training recipe or covariate contract.	Single expert, fixed-fusion component, router expert, and KD anchor. Not claimed as an exact replay of an external CAPER release or as a generally defined architecture.
Compact TimeXer-168	Local compact TimeXer-family artifact; 2,358,788 parameters in the latency serving artifact. Exact upstream repository revision for the UrbanEV artifact: unknown/not claimed .	Existing UrbanEV predictions with 168-hour context and shared target identity. No new expert training was performed for the audit chain.	Best evaluated single UrbanEV model within the frozen candidate set. Not “official upstream TimeXer”, universal best, or SOTA.
Global Fixed	Deterministic convex combination $\alpha_f \hat{y}^T + (1 - \alpha_f) \hat{y}^C$; one fold-level scalar pooled across horizons. Serving count 2,406,781 parameters includes both experts and scalar.	Fold 1 uses paired pre-test validation; folds 2–6 use earlier-fold OOF predictions only. Current-fold targets cannot fit α_f .	Selected fixed reference and distillation teacher. Not a cost-free ensemble, blind-test result, or cross-dataset claim.
Router v1	Local CAPER_TimeXer_ReliabilityGate; direct-loss MLP has hidden widths 32/16, horizon embedding 4, dropout 0.1, Huber loss, seed 42. Authoritative deployed parameter count: unknown/not claimed .	Expert outputs/disagreement plus 21 forecast-origin state features, rolling cross-fit; no feature timestamp exceeds the forecast origin.	Frozen learned-router comparison. It failed the 1% gate; no robustness or deployment qualification is claimed.
Chronos-2	Official amazon/chronos-2 snapshot resolved to commit 29ec3766d36d...e8498c; chronos-forecasting 2.2.2, Torch 2.5.1+cu121. Parameter count: unknown/not claimed .	Zero-shot FP32; 168-hour multivariate context; native Q0.5 output; no future covariates; predictions clipped to $[0, 1]$ for the frozen primary metric.	One teacher-qualification screen. Its failure is not evidence that foundation models generally fail for EV forecasting.
Residual student / aligned KD	Local CAPER-anchor plus shared DLinear residual branch; 338 trainable parameters. GT-only, aligned-KD, and shuffled-teacher variants share architecture, initialization, optimizer, data, epochs, and seed.	Occupancy-rate-only input, 168-hour lookback, 275 channels; aligned loss adds the frozen teacher-residual term with $\lambda = 0.5$.	Negative-gated distillation experiment. The aligned-vs-shuffled subcheck is not a passed improvement over GT-only and does not authorize latency or architecture search.
Paris CAPER	Paris_CAPER_phase_only, newly trained local adaptation; no UrbanEV checkpoint or fitted coefficient loaded. Authoritative parameter count: unknown/not claimed .	Development-only. Available, Charging, Passive, and Other map to available/active/unavailable; daily/weekly ordinal lags; target-history encoder plus station embedding replaces unavailable UrbanEV channels.	Paris single-model candidate. Must not be named unqualified CAPER or treated as an exact UrbanEV replay.
Paris TimeXer	TimeXer_local_audited_compact_L1 68; local source SHA-256 begins 563d4ce4304f; model class source matches the workspace copy. Upstream commit is unknown/not claimed .	Retrained from initialization on Paris development; compact profile ($d = 128$, four heads, two layers); external train-only scaler, masked loss, direct endpoint, no time markers.	Paris single-model candidate. Explicitly a local audited adaptation, not an exact official upstream implementation.
Paris strict teacher	Fold-specific convex combination of the two Paris local adaptations; no separately trained joint model and no protected-target input.	Fold 1 α from pre-evaluation validation; later folds from prior-fold OOF predictions; shared station, mask, target, and timestamp identities.	Development-only teacher qualification. It failed the point-effect gate; student training and formal/protected evaluation were not authorized.

3 Research questions and supporting results

Table 4: Research-question map from operational definition to bounded answer.

RQ	Operational question	Evidence and analysis	Direct result	Bounded answer
RQ1	How many frozen configurations are exact, disclosure-limited, trend-only, underidentified, failed, or non-complete?	1,224-row status registry; deterministic S1–S6 classifier; execution state, source-cell mapping, fold coverage, and worst disclosed-metric grade.	865 artifacts complete; S1=156, S2=216, S3=122, S4=371, S5=19, S6=340.	Completion is broader than exact reproduction in this inventory. Counts are not a field-wide reproducibility estimate or a model ranking.
RQ2	How much oracle complementarity survives feasible prequential fusion and learned routing?	Target-dependent oracle; expert residual diagnostics; forecast-origin proxy classifier; strict Global/Horizon Fixed controls; direct-loss and secondary router variants.	Oracle +16.926% versus TimeXer; Global Fixed +3.178%; primary router +0.443% versus Global Fixed.	Global Fixed retains a 3.178% internal point gain. The smaller router gain fails the frozen effect gate.
RQ3	What changes after synchronous latency and the minimum-effect rule are enforced?	RTX 4060 FP32 batch 1/8/32 benchmark; batch-1 primary endpoint; router, Chronos-2, KD, and Paris conjunctive gates.	Global Fixed has +48.86% batch-1 P50 versus TimeXer; later gains are +0.443%, −3.048%, +0.119%, and +0.847%.	The accuracy leader has a measured serving cost; positive or consistent subchecks do not automatically qualify the next engineering stage.
RQ4	How do artifact checks, separate storage, and stopping rules affect which claims are supported?	Metric lock; non-selective control rerun; 32/32 Paris prediction closure; unchanged-prediction repair; versioned termination; procedural vault guard.	Three identity shuffles were excluded before metrics; 10,236 masked target entries were restored without prediction changes; analytical access after the Paris lock remained zero.	The checks blocked invalid results and stopped failed versions. They do not prove higher accuracy, hardened security, or external validity.

A Reproducibility and Audit Appendix

This appendix documents the audit checks used in the case study. It reuses frozen main-workspace artifacts and performs no model training or inference. It does not recompute any experiment metric, run a new bootstrap or threshold sensitivity, or read Paris formal or protected target content. The tables separate facts stored in the artifacts from suggestions that may be useful elsewhere. When a field cannot be established from a frozen artifact, it is reported as *unknown/not claimed* rather than filled in by assumption.

The appendix does not add analyses that were not run. There was no independent-coder classification exercise, inter-rater agreement statistic, alternative bootstrap block-length analysis, or new Global Fixed uncertainty interval. The 0.5/1/2% comparison is arithmetic context only; it does not recompute metrics or change a decision. The S1–S6 registry is a deterministic single-pipeline classification of one inventory. Historical project records used a review label for what was actually a separately instantiated automated differential artifact audit, not a blinded external human replication or an inter-rater study.

A.1 Deterministic S1–S6 decision tree

The classifier in `experiments/00_registry_reclassification/scripts/rebuild_status_registry.py` applies the following ordered rules to each frozen manifest row. Order is part of the definition; a later numerical rule cannot overwrite an earlier execution fact.

- D1.** If `execution_status=failed`, assign S5 and require any future attempt to use a new versioned retry identifier.
- D2.** Otherwise, if `execution_status` is not `complete`, assign S6 and require any future execution to use a new versioned run identifier.
- D3.** Otherwise, if the row maps to UrbanEV source Table 4, assign S4 because the source paper and repository do not identify the disclosed node uniquely.
- D4.** Otherwise, if no source comparison cell can be mapped uniquely, assign S4.
- D5.** Otherwise, collect the disclosed RMSE, MAPE, RAE, and MAE grades and take the worst valid grade. The frozen bands are exact (relative error no greater than 2%), near (2–5%), and trend (greater than 5%); an empty valid-grade set defaults to trend when a comparison record exists.
- D6.** If comparison fold coverage is below six, assign S3. Otherwise, if the worst grade is trend, assign S3.
- D7.** Otherwise, if the track is `official` and the worst grade is exact, assign S1. Every other completed, uniquely mapped, non-trend row is S2.

The rules therefore apply in this order: S5/S6 → S4 → S1/S2/S3. A completion mark indicates process completion only; it is not a synonym for S1. The following examples are the first row of each final status in the authoritative CSV. The `classification_reason` text is an English rendering of the Chinese source field so that the appendix compiles under the paper’s Latin-font setup.

Table 5: One frozen registry example per evidence status. Full run identifiers and reasons are retained to make the classification path inspectable.

State	Example ID	Reason
S1	official-compat-v1-t3-lo-occ-3-1-region-None-42-20-eae54b2d	Official-compatible protocol is identifiable, all six folds are complete, and the relative error of each of four disclosed metrics is no greater than 2%.
S2	audited-t3-lo-occ-3-1-region-None-42-20-efalea0d	Six folds and the main protocol align, but an audited correction is present or at least one metric lies in the 2–5% approximation band.
S3	audited-t3-ar-occ-3-1-region-None-42-20-3b0793b0	At least one paper-disclosed metric has relative error above 5%; only trend-level evidence is retained.
S4	audited-t4-lo-occ-3-1-194-None-42-20-42da10db	Neither the paper nor repository discloses the Table 4 node identity; node 194 is a predeclared reverse-identification candidate, not a unique truth.
S5	audited-t3-lstm-occ-6-3-region-None-42-20-c735895e	The frozen configuration was executed but failed; the failure artifact is retained and cannot contribute a paper number.
S6	audited-t3-TimesNet-occ-3-3-region-None-42-1-1687eb9d	The configuration is frozen in the manifest but was not executed or registered after accumulated failures in the same model batch.

A.2 Model identity is a claim field

A model name alone does not identify the exact implementation. The same family label may refer to an upstream release, a local source snapshot, an existing prediction artifact, a retrained adaptation, or a fixed combination of other artifacts. Table 3 reports the identity recorded in the frozen artifacts and lists fields that remain unknown. This is especially important for Paris: `Paris_CAPER_phase_only` and `TimeXer_local_audited_compact_L168` are development-only local adaptations, not exact transfers of the UrbanEV experts or claims about every upstream implementation.

A.3 Fail-closed state machine

The audit states are:

REGISTERED → RUNNING → CLOSED_UNVERIFIED → AUDIT_PASS → METRIC_UNLOCKED → GATE_PASS or
GATE_FAIL.

Three rules prevent a defective artifact from moving forward. First, an incomplete matrix, target/order mismatch, invalid control, boundary violation, or provenance failure moves `CLOSED_UNVERIFIED` to `BLOCKED` or `REPAIR_REQUIRED`; aggregate metrics remain inaccessible. Second, a permitted repair creates `REPAIRED_UNVERIFIED`, records parent and child hashes and whether predictions changed, and returns to the full audit. It does not move directly to metric unlock. If selective repair could use information about which controls failed, the full control family is invalidated and rerun. Third, `GATE_FAIL` is terminal for the current protocol version. Continuing with a changed model, threshold, loss, or evidence role requires a new version and a still-valid data boundary.

Two incidents show how these transitions worked. In M7-C, the first audit detected three identity permutations at 69/72 cells while `metrics_unlocked=false`. All 24 shuffled controls were invalidated and rerun before the 72/72 audit passed. In Paris, 32 prediction jobs closed without evaluation metrics. Missing-target representation was repaired in 24 artifacts (10,236 entries) without changing predictions, and development metrics were unlocked only after the 32/32 prediction audit passed. These are integrity events, not accuracy improvements.

A.4 Common lock schema and invariants

The project used several lock and receipt formats rather than one universal schema. Table 6 gives a common interface for the rules that connect each claim to its evidence. It does not imply that every historical artifact contains every field. A missing field must be recorded as `unknown` or fail the relevant gate; it must not be inferred from a filename.

Table 6: Normalized lock schema. Existing examples use different field names; the rightmost column states the rule an implementation must enforce.

Group	Required fields	Existing artifact exemplar	Invariant
Protocol	<code>protocol_id</code> , version, status, protocol SHA-256	router, latency, distillation, and Paris teacher protocol files	Data role, endpoint, reference, gate, and stop rule are immutable before aggregate access.
Configuration	<code>run_id</code> , config fingerprint, model identity, seed, fold, horizon	status registry and run manifests	One identifier maps to one configuration; retries retain parentage rather than overwriting identity.
Impl.	code/source hashes; dependency and bundle hashes	Chronos model lock; Paris implementation fingerprints	The implementation that emitted predictions equals the registered implementation or has an explicit non-training amendment receipt.
Environment	Python and library versions; accelerator, runtime, precision; optional environment hash	latency hardware/software manifests; Chronos environment block	Scope is explicit. If no unified environment hash exists, reproducibility is limited rather than implied.
Data boundary	source/data hash, role, permitted metadata, fold dates	baseline protocol; Paris boundary lock and source manifest	Development, formal, and protected roles cannot be silently exchanged.
Target identity	target-value, index/order, time, spatial-unit, and mask hashes; fold-horizon cell prediction SHA-256, shape, finiteness/range, attempt path	expert diagnostics, run status, prediction audits	Candidate and reference use identical target positions and semantics for every compared cell.
Prediction		per-run receipts and prediction audits	The audited bytes are the bytes aggregated; a repair changing predictions requires new scientific evidence.
Metric	metric script/hash, endpoint definition, output hash	metric-lock scripts, aggregate summaries	The metric implementation and weighting rule are fixed; output cannot precede unlock.
Unlock	lock time, conditions, authority, receipt SHA-256, matrix/state hash	Paris metrics-unlock receipt; M7-C formal audit	Every condition is conjunctive and machine-checkable; role-specific unlock does not unlock another role.
Access	analytical-access counter, operation definition, log/receipt location	Paris protected guard and prediction audit	Counter semantics and time origin are stated; storage migration is recorded separately.
Repair	parent hash, child hash, reason, prediction-changed flag, audit return state	shuffled-control amendment and masked-target repair ledger	Repair is traceable and returns to unverified state; non-selective scope is used where observation could bias repair.
Decision	reference, point effect, interval, fold/cell/horizon checks, gate vector, authorized next action	router, Chronos, D1, and Paris teacher summaries/decisions	A failed required component yields terminal failure for that version even when another subcheck passes.
Claim	claim ID, authoritative artifact/hash, evaluation type, allowed/prohibited wording, scope limit	Table 7	Manuscript wording cannot exceed the evidence type or data role.

A.5 Operational definition of analytical access

The Paris guards distinguish three operations.

Storage materialization. Copying, moving, splitting, preserving, or hashing target-bearing files without using their values. Creating or checking a path/size/hash manifest also belongs to this category. These operations are recorded as storage events and do not count as target analysis.

Approved metadata handling. Reading only the registered non-target fields (date boundary, station identifier, latitude, longitude, postcode, and area), row counts, file sizes, and hashes needed to establish the split and storage receipt. State and target values are not used.

Analytical access. Any post-lock use of formal or protected target values. This includes displaying, summarizing, plotting, filtering by, learning from, scoring against, or making a scientific claim from those values. Manual viewing also counts. Use in training, validation, model selection, metrics, threshold selection, error analysis, or a paper table counts even when no file is modified.

Under this definition, `analytical_access_count=0` means that no protected label was used after the boundary lock for the listed analytical operations. It does not mean that target bytes were never materialized: one storage migration created vault shards and moved recoverable sources. It also does not erase the logged display of two example rows during pre-lock schema discovery. The counter is supported by project receipts and audits. It is not enforced by the operating system, does not provide encryption, and cannot prevent deliberate bypass by an authorized user.

These controls are intended to prevent accidental use, or use chosen after earlier results were seen, within the project workflow. Examples include a script path that reaches a future role, a stale cache, recoverable Git objects, or a metric opened before the registered closure. The vault path, manifest, hashes, guards, and negative main-project scan reduce these risks. Operating-system compromise, deliberate bypass, encryption, and immutable access logging are outside scope. The supported claim is source separation and zero recorded analytical use, not hardened security.

A.6 Paris fold-synchronous bootstrap, including the partial block

The frozen implementation in `experiments/09_distill_v2/scripts/aggregate_paris_teacher.py` used 5,000 replicates and seed 42. For each development fold, it found target timestamps shared by all four horizons and sorted them. It required one contiguous hourly sequence and divided that sequence at offsets 0, 24, 48, and so on. The final slice was kept even when it contained fewer than 24 hours.

Each replicate sampled the fold's original number of blocks with replacement. The same sampled timestamp sequence was used for every horizon in that fold. Observed-target masks were then applied within each cell, and cell RMSEs were macro-averaged. The better CAPER/TimeXer reference was selected again inside the replicate, and teacher gain was computed against that reference. The reported interval uses the 2.5th and 97.5th percentiles.

August and October each contain 732 common hourly timestamps: 30 full 24-hour blocks and one 12-hour block (31 selectable blocks). September and November each contain 708 timestamps: 29 full blocks and one 12-hour block (30 selectable blocks). Every block, including the 12-hour partial block, had the same probability of selection. Because the partial block contained fewer timestamps, replicate length varied with the number of times it was drawn. No padding, wraparound, discarding, or extra weighting was applied. The reported Paris interval applies to this exact rule. No alternative block length or partial-block sensitivity analysis was run.

A.7 Claim-to-artifact links

Table 7 links each headline result to the machine-readable artifact that supports it. The hash is computed over the named artifact in the main workspace. A hash identifies file bytes; it does not prove that the method is correct. The evaluation-type and wording columns state the claim limit. For example, the Global Fixed row supports an internal point comparison and a separate measured latency result. It does not gain an unreported bootstrap interval merely because later stages used one.

Table 7: Claim-to-artifact trace and wording guardrails (part 1 of 2). Hashes are the first 12 hexadecimal characters of SHA-256 over the named frozen artifact.

ID	Headline result	Authoritative artifact / SHA-256 ₁₂	Evaluation type	Allowed wording; prohibited wording
C1	Frozen registry: 1,224 configurations; 865 complete artifacts; S1=156.	experiments/00_registry_reclassification/outputs/CONFIG_STATUS_SUMMARY.json 833352c136a4	registry audit	<i>Allowed:</i> In this inventory, completion and exact reproduction differ. <i>Prohibited:</i> A field-wide reproducibility rate or a model-quality ranking.
C2	Source-table status distributions are uneven under the frozen mapping.	experiments/00_registry_reclassification/outputs/CONFIG_STATUS.csv 4b61c2aea510	registry audit	<i>Allowed:</i> Report S1-S6 counts by UrbanEV source table. <i>Prohibited:</i> Call an S4 model invalid or intrinsically weak.
C3	Target-dependent CAPER-TimeXer oracle gain is 16.926%.	experiments/01_expert_diagnostics/outputs/EXPERT_DIAGNOSTIC_TIC_SUMMARY.json 53fa7f22fca6	real-GT oracle diagnostic	<i>Allowed:</i> An infeasible upper bound indicates local complementarity. <i>Prohibited:</i> The oracle is deployable or its gain is recoverable.
C4	State-feature expert-win AUC is 0.6151.	experiments/01_expert_diagnostics/outputs/STATE_FEATURE_DECISION.json 9f84ac483e02	self-supervised proxy	<i>Allowed:</i> Relative expert advantage is moderately predictable. <i>Prohibited:</i> AUC proves lower forecast RMSE or causal demand knowledge.
C5	Strict Global Fixed RMSE is 0.071367, a 3.178% internal gain.	experiments/03_fixed_fusions/outputs/FIXED_FUSION_SUMMARY.json 0a0162204c83	internal rolling real GT	<i>Allowed:</i> Improves on TimeXer under the audited UrbanEV protocol. <i>Prohibited:</i> Blind-test, cross-city, efficient, or deployment-ready.
C6	Direct-loss router gains 0.443% and fails the frozen 1% gate.	experiments/05_router_v1/outputs/ROUTER_V1_SUMMARY.json a9dfc2ebe4a6	internal real GT; gated	<i>Allowed:</i> A positive sub-threshold internal signal; current stage stopped. <i>Prohibited:</i> Router qualification, robustness, or post-hoc gate revision.
C7	Global Fixed batch-1 P50 is 48.86% above TimeXer.	experiments/06_static_latency/LATENCY_SUMMARY_TABLE.csv f94d0a7a9e95	hardware-specific latency	<i>Allowed:</i> Measured on the frozen RTX 4060 FP32 sequential workload. <i>Prohibited:</i> Portable latency, low cost, or universal serving efficiency.

Table 8: Claim-to-artifact trace and wording guardrails (part 2 of 2). Hashes are the first 12 hexadecimal characters of SHA-256 over the named frozen artifact.

ID	Headline result	Authoritative artifact / SHA-256 ₁₂	Evaluation type	Allowed wording; prohibited wording
C8	Chronos-2 gains -3.048% versus Global Fixed and fails qualification.	experiments/06_5_teacher_qualification/outputs/CHRONOS2_QUALIFICATION_SUMMARY.json 01e8084c16ae	zero-shot internal screen	<i>Allowed:</i> This locked revision does not qualify for this protocol. <i>Prohibited:</i> Foundation models are generally inferior for EV data.
C9	Aligned KD gains 0.119% versus identical GT-only; CI crosses zero.	experiments/07_residual_distillation/outputs/D1_SUMMARY.json c38163157c89	internal real GT; gated	<i>Allowed:</i> The primary distillation comparison fails; distill_v1 stops. <i>Prohibited:</i> Distillation improves the student or merits D2 latency.
C10	Aligned KD gains 0.259% versus shuffled teacher.	experiments/07_residual_distillation/outputs/D1_SUMMARY.json c38163157c89	negative control	<i>Allowed:</i> Evidence of a small sample-aligned teacher signal. <i>Prohibited:</i> Engineering value or a passed primary KD claim.
C11	Paris teacher gains 0.847%, wins 4/4 folds and 16/16 cells, but fails 1%.	experiments/09_distill_v2/outputs/teacher_qualification/aggregate/PARIS_TEACHER_QUALIFICATION_SUMMARY.json 5b7c1ddc353c	development real GT	<i>Allowed:</i> Consistent development gain below the frozen qualification effect. <i>Prohibited:</i> Round to 1%, imply formal/protected performance, or call it official TimeXer.
C12	Three identity shuffles were blocked at 69/72 before metric unlock.	experiments/07_residual_distillation/outputs/FORMAL_MATR_IX_AUDIT_V1_0_FAIL_IDENTITY_SHUFFLE.json 584313534390	artifact integrity	<i>Allowed:</i> The lock forced a non-selective rerun of all 24 controls. <i>Prohibited:</i> Use the invalid controls as performance evidence.
C13	Paris prediction audit passed 32/32 before development-metric unlock.	experiments/09_distill_v2/outputs/teacher_qualification/PARIS_TEACHER_PREDICTION_AUDIT.json ee882d8e907b	artifact integrity	<i>Allowed:</i> Identity checks authorized development-metric aggregation. <i>Prohibited:</i> The audit proves model correctness or improved RMSE.
C14	Protected analytical-access counter is 0; vault materialization is true.	experiments/08_paris_new_evidence/PROTECTED_ACCESS_GUARD.json a7780e39f4ef	procedural data boundary	<i>Allowed:</i> No protected label was used analytically after the boundary lock. <i>Prohibited:</i> Targets never existed, were never materialized, or are cryptographically inaccessible.

A.8 Latency across the frozen batch grid

The main text uses batch-one end-to-end P50 as its primary engineering endpoint. Table 9 lists the complete batch 1/8/32 grid from the frozen latency summary. The script `experiments/10_audit_paper/scripts/generate_revision_appendix.py` checks all 18 method–batch rows and formats the stored values without rerunning the benchmark.

The hardware and software settings were Windows 10 build 26200, 16 physical/32 logical AMD CPU cores, approximately 16.8 GB RAM, an RTX 4060 Laptop GPU with driver 595.71, Python 3.10.8, Torch 2.5.1+cu121, CUDA 12.1, cuDNN 9.1, FP32, and single-threaded Torch/OMP/MKL host settings. GPU clocks were observed per process rather than pinned. Model loading and checkpoint I/O were excluded. Cold-start, quantized, compiled, parallel-expert, and serverless settings were not tested.

Table 9: Frozen end-to-end synchronous latency at batch sizes 1, 8, and 32. P50 and P95 are milliseconds per request; throughput is forecast origins per second at P50. Every value is formatted directly from `LATENCY_SUMMARY_TABLE.csv`; no benchmark was rerun.

Method	Batch origins	E2E P50 (ms)	E2E P95 (ms)	Origins s^{-1} (P50)
DLinear	1	1.166	1.763	857.9
DLinear	8	2.118	2.874	3776.9
DLinear	32	5.382	7.728	5945.7
NLinear	1	0.674	1.056	1483.7
NLinear	8	1.554	2.101	5148.0
NLinear	32	3.896	5.373	8213.7
GBDT-Forecaster	1	6.242	7.195	160.2
GBDT-Forecaster	8	40.188	50.261	199.1
GBDT-Forecaster	32	157.087	186.205	203.7
CAPER	1	7.934	10.798	126.0
CAPER	8	10.740	14.253	744.9
CAPER	32	19.753	24.266	1620.0
TimeXer	1	14.067	15.952	71.1
TimeXer	8	105.780	113.816	75.6
TimeXer	32	482.165	519.827	66.4
Global Fixed CAPER+TimeXer	1	20.940	33.123	47.8
Global Fixed CAPER+TimeXer	8	125.350	136.947	63.8
Global Fixed CAPER+TimeXer	32	479.634	510.574	66.7

Workload: one request produces all four horizons for 275 UrbanEV zones. Measurements used FP32, one RTX 4060 Laptop GPU, sequential expert execution, five processes, 50 warm-ups, and 1,000 timed iterations per process and method–batch cell. Throughput is not monotone across model families because the frozen serving graphs batch differently; these values should not be assumed to hold in other deployment settings.

A.9 Protocol and environment artifact index

The following index lists existing controls. Listing a file does not make missing environment information complete. The latency and Chronos stages have explicit environment records. Other stages retain protocols, source/configuration fingerprints, and run receipts but do not provide one cross-stage environment hash. Exact environment equivalence is therefore unknown/not claimed where that record is absent.

Table 10: Protocol and environment artifact index (part 1 of 2).

Artifact	Function	Scope limit
experiments/00_registry_reclassification/scripts/rebuild_status_registry.py experiments/00_registry_reclassification/outputs/CONFIG_STATUS.csv experiments/02_lightweight_baselines/configs/BASELINE_PROTOCOL.json refine-logs/ROUTER_PROTOCOL_V1_0.md and experiments/05_router_v1/configs/ROUTER_V1.json experiments/06_static_latency/LATENCY_PROTOCOL_V1_0.md	Deterministic S1–S6 implementation and priority order Row-level run IDs, fingerprints, grades, reasons, and rerun policy UrbanEV target, folds, horizons, seed, and candidate protocol Router feature/training configuration and frozen Gate 4 Request shape, synchronization, warm-up, repetitions, and primary endpoint	Single programmed classifier; no inter-rater validation One frozen UrbanEV inventory Candidate-set internal evaluation only One seed; proxy and real-GT roles remain separate Warm synchronous serving, not cold start
experiments/06_static_latency/hardware_manifest.json and experiments/06_static_latency/software_manifest.json experiments/06_static_latency/LATENCY_SUMMARY_TABLE.csv	Device, driver, platform, runtime, libraries, precision, host threads Batch 1/8/32 model-only and end-to-end distributions, throughput, memory	One laptop system; clocks observed rather than pinned Hardware/workload-specific
experiments/06_5_teacher_qualification/TEACHER_QUALIFICATION_PROTOCOL_V1_0.md experiments/06_5_teacher_qualification/CHRONOS2_MODEL_LOCK.json experiments/07_residual_distillation/DISTILL_PROTOCOL_V1_0.md experiments/07_residual_distillation/configs/FORMAL_STUDENT_CONFIG_V1.json experiments/07_residual_distillation/SHUFFLE_CONTROL_SPEC.json	Chronos-2 qualification endpoint and gate Resolved model commit, file hashes, snapshot bytes, and environment Teacher, student, loss, controls, milestones, stop rules 338-parameter student input/architecture/training identity 24-hour block-shuffle negative-control construction	Exact project protocol only One revision and zero-shot profile Distill v1 only; later search prohibited after failure One architecture and seed Control test, not a second primary endpoint

Table 11: Protocol and environment artifact index (part 2 of 2).

Artifact	Function	Scope limit
experiments/07_residual_distillation/outputs/FORMAL_MATRIX_AUDIT_V1_0_FAIL_IDENTITY_SHUFFLE.json and experiments/07_residual_distillation/outputs/FORMAL_MATRIX_AUDIT.json experiments/07_residual_distillation/outputs/D1_SUMMARY.json and experiments/07_residual_distillation/outputs/D1_DECISION.md experiments/08_paris_new_evidence/DATA_BOUNDARY_LOCK.json and experiments/08_paris_new_evidence/PROTECTED_ACCESS_GUARD.json experiments/08_paris_new_evidence/PARIS_EVIDENCE_VAULT_REPORT.md experiments/09_distill_v2/PARIS_TEACHER_QUALIFICATION_PROTOCOL_V1_0.md experiments/09_distill_v2/PARIS_TEACHER_MODEL_CONFIG.json experiments/09_distill_v2/TIMEXER_LOCAL_ADAPTATION_PROVENANCE.json and experiments/09_distill_v2/PARIS_CAPER_ADAPTATION_V1_0.md experiments/09_distill_v2/PARIS_TEACHER_DEPLOY_APPROVAL_V1_1.md experiments/09_distill_v2/outputs/teacher_qualification/PARIS_TEACHER_PREDICTION_AUDIT.json and experiments/09_distill_v2/outputs/teacher_qualification/PARIS_TEACHER_METRICS_UNLOCK_RECEIPT.json experiments/09_distill_v2/outputs/teacher_qualification/aggregate/PARIS_TEACHER_QUALIFICATION_SUMMARY.json experiments/09_distill_v2/PARIS_TEACHER_RESULT_REVIEW.md experiments/10_audit_paper/scripts/generate_revision_appendix.py	Pre-unlock failure and final 72/72 audit Primary/negative-control effects and terminal gate decision Role dates, permitted metadata, storage state, access counter, unlock rule Vault identities, hashes, negative main-project scan, threat limits Development folds, target, missingness, time rule, teacher fit, gate Local CAPER/TimeXer profiles, training, queue, bootstrap Source hash and explicit local adaptation differences Automated differential audit closure after the earlier block 32/32 closure, identities, role-specific metric unlock, access counts Point effect, fold/cell results, synchronized bootstrap, gate vector Separately instantiated result-to-claim check and provenance warnings Mechanical hash/latency table generation	Supports control integrity, not accuracy No D2 latency, multi-seed, or protected evidence Procedural guard; not hardened authorization Metadata/hash audit only; no content-level protected audit Paris development only Local adaptations, not exact upstream implementations Upstream TimeXer commit undisclosed; Paris CAPER newly trained Automated artifact audit, not blinded human replication Development-metric authorization only Current hypothesis failed; no student/formal/protected result Warnings limit stronger process claims; they do not reverse the gate No model, metric, resampling, or vault operation

A.10 Author and reviewer application checklist

Table 12 summarizes checks that authors and reviewers can apply. Missing validation must be stated as missing. The 1% rule is a local convention without an economic utility model. The S1–S6 labels come from one deterministic classifier and have no inter-rater evidence. The model and latency conclusions are limited to their recorded seeds, panels, adaptations, and serving setup.

Table 12: Author and reviewer checklist derived from this study.

ID	Check	Author must expose	Reviewer test
1	Target and role semantics	numerator/denominator, spatial unit, mask, sampling clock, and development/formal/protected role	Could two datasets or target constructions be mistaken for the same RMSE scale?
2	Configuration identity	run ID, protocol version, fingerprint, source mapping, seed, fold/horizon, and execution state	Could a completed run still be underidentified or map to several disclosed cells?
3	Information timing	forecast origin, lookback, covariates, fill/scaler fit interval, and prequential weight/tuning source	Does any current/future target influence a parameter evaluated in the same fold?
4	Artifact closure	expected matrix cardinality, terminal attempts, prediction/target/order/time/spatial/mask identities	Are aggregates withheld until every required cell is closed and verified?
5	Metric lock	metric implementation identity, lock state, unlock conditions, authority, receipt, and access counter	Can any result be read or selected before the audit passes?
6	Comparator and endpoint	primary reference, macro/micro weighting, endpoint, sign convention, and secondary diagnostics	Was the reference or endpoint changed after seeing results?
7	Uncertainty and gate	resampling unit, partial-block rule, replicates/seed, minimum effect, consistency checks, stop rule	Is a positive interval against zero being substituted for a failed effect-size requirement?
8	Controls and repair	negative-control construction, invalidation scope, before/after hashes, and whether predictions changed	Were only visibly bad cells repaired, or was the formal control family rerun non-selectively?
9	Model identity	upstream/local source, checkpoint/revision, local adaptations, input contract, and every unknown field	Does a local adaptation masquerade as an official or exact upstream implementation?
10	Engineering measurement	device, software, precision, batch/request shape, synchronization, warm-up, repetitions, P50/P95, throughput	Is a hardware-specific measurement being presented as general efficiency?
11	Future/protected evidence	storage versus analytical access, allowed metadata, threat-model limit, logs, and unlock rule	Does the boundary prevent analysis, merely record storage, or claim stronger security than implemented?
12	Claim source	claim ID, authoritative artifact, SHA-256, evaluation type, allowed/prohibited wording, scope limit	Can every headline number be traced, and are diagnostics/subchecks kept at their original evidence type?
13	Negative result	terminal decision, disallowed same-interval actions, and requirements for a new version/evidence boundary	Was a failed hypothesis silently continued through tuning, threshold changes, or a new structure?
14	Missing validation	limitations must name absent multi-seed, multi-hardware, external-team, utility, or rater evidence	Is any absent analysis implied by rhetoric such as robust, validated, secure, or general?

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